

# ACE-CAN attack-centric CAN IDS pipeline

Repair the objective first, then test feature-preserving detection.

## Input conditions

### I1 Rare traces

CAN ID, DLC, payload  
timestamp, source  
attack rate < 1%

### I2 Cross-shift

unknown vehicle  
unknown attack family  
normal traffic dominates

## Branch operation

### E1 Metric audit

all-normal check  
weighted-score ambiguity  
ranking inversion

### D1 GRAIN-CAN

same-ID timing  
payload dynamics  
ID behavior

## Evidence outputs

### E2 Corrected eval

attack P/R/F1  
AUPR, R@FPR  
event recall

### D2 Claim boundary

strong baseline  
low-FPR evidence  
not solved

metric trap:  
high score, zero recall

Attack-F1 AUPR R@FPR Event

Output: corrected benchmark + deployment evidence + reproducible reporting checklist